Broader coverage changes the findings

Four image models; provisional unattended labels; weak anomaly diagnostics; retained smoking/VadCLIP

55,156

New GPU predictions

40 completed model/split runs

1,770

Unattended endpoints

36 previously; 49.2× increase

962

Held-out inputs / protocol

242 positive + 720 normal

MiniCPM: strongest raw frame F1

F1 0.479 • recall 34.7% • FPR 3.47%
Many positive endpoints are still missed.

Three anomalies now attempted

6,667 littering; 1,675 loitering; 111 pose scenes.
Weak labels limit accuracy claims.

Use the available data at its valid level

13,789 input/protocol evaluations per model; shared images and endpoints are counted explicitly

Study	Dev	Test	What the labels support
Unattended: frames	808	962	Provisional event/control comparison
Unattended: 20 s windows	808	962	Same endpoints, four-frame sheets
Littering candidates	6,667	—	Weak action labels; target-prediction rate
Loitering candidates	1,675	—	Person boxes; duration unverified
Person presence	1,675	—	Same images; grounding diagnostic
Rough-sleeping candidates	111	—	Synthetic lying/fall scenes
Lying vs standing crops	73	48	Oracle boxes; posture diagnostic

More sampled inputs do not imply more independent events or cameras.

Recover events from source timelines

Assistant review precedes model inference; independent human verification remains necessary

1. Inspect source videos

Chronological review boards
~2 fps native frames
CAVIAR role/XML checks

18 included recordings; raw/additional duplicates removed.

2. Freeze intervals

Choose visible departure and clear normal controls.

Exclude occlusion, nearby owners and items lacking placement history.

3. Record provenance

Source SHA and frame index
Reviewed interval decisions
Decoded image identity

No inferred labels from model responses.

Labels are provisional assistant review. These results are an event study, not a verified deployment benchmark.

Related camera recordings stay together

Camera-family holdout avoids mixing adjacent views and recordings between development and test

Development: 3 camera families

CAVIAR: INRIA
ABODA: lobby + entrance

808 endpoints: 255 positive / 553 normal
Fit CLIP thresholds on these inputs.

Test: 2 held-out families

ABODA: walkway + classroom
7 recordings; 5 positive intervals

962 endpoints: 242 positive / 720 normal
Apply frozen thresholds once.

The unit of independence is the source/camera group. 962 adjacent endpoints are not 962 independent events.

Run quality evaluation without pacing

RTX 5000 Ada; four isolated model workers; all eligible inputs; no additional latency pilot

Model	CUDA dtype	Batch	Evaluation input
MobileCLIP2-S2	float32	64	Shared image / contact sheet
SigLIP2-B/16-224	float32	64	Shared image / contact sheet
SmolVLM-500M	bfloat16	1	Deterministic 64-token reply
MiniCPM-V 4.0	bfloat16	1	64-token reply; one slice

Shared preprocessing: letterbox, maximum side 768 px. Windows: four chronological frames spanning 20 seconds, with no future context. Saved revisions and raw outputs make the comparison reviewable.

No pacing sleeps or accuracy RAM cap; concurrent runs are not a fair latency comparison.

Keep event quality and diagnostics separate

Abstentions remain in positive denominators; score ranking is separate from categorical decisions

Reviewed event comparison

Precision / recall / F1 / FPR
CLIP score AP / AUROC

Frozen development thresholds
Camera-group uncertainty

Weak candidate diagnostics

Target-prediction rate
Abstentions + errors

No event precision or FPR
No verified anomaly recall

Oracle pose component

Lying versus standing crops
Source-image split

Ground-truth boxes are free
Only four test negatives

All 55,156 saved predictions were checked; recomputed metrics match all 40 summaries.

MiniCPM has the highest raw frame F1

Provisional held-out labels: 242 positive + 720 normal endpoints; two camera families

Model	Precision	Recall	F1	FPR	AP
MobileCLIP2-S2	25.31%	75.21%	0.379	74.58%	0.209
SigLIP2-B/16-224	0.00%	0.00%	0.000	2.64%	0.309
SmolVLM-500M	18.58%	63.64%	0.288	93.75%	—
MiniCPM-V 4.0	77.06%	34.71%	0.479	3.47%	—

MiniCPM: 84 TP / 25 FP / 158 FN / 695 TN. Higher-recall alternatives also flag most normal endpoints. SigLIP score ranking exceeds its raw decision quality.

AP reference prevalence: 0.252. VLM AP is unavailable. Raw frame response coverage is 100%.

More temporal context has mixed effects

Same 962 endpoint labels; four-frame sheets cover the preceding 20 seconds

Model	Precision	Recall	F1	FPR	AP
MobileCLIP2-S2	27.18%	76.03%	0.400	68.47%	0.362
SigLIP2-B/16-224	12.50%	1.65%	0.029	3.89%	0.325
SmolVLM-500M	24.95%	96.28%	0.396	97.36%	—
MiniCPM-V 4.0	96.30%	10.74%	0.193	0.14%	—

MiniCPM trades recall for fewer false alarms. SmolVLM alarms on almost every input (14 unanswered). The old 6-second sheets and new 20-second sheets are different protocols.

Development false-alarm budgets transfer poorly

Thresholds fitted on 553 normal + 255 positive development endpoints; frozen for test

Input	Model	Dev selection	Threshold	Recall	FPR	F1
frame	MobileCLIP	Max dev F1	0.247	100.00%	100.00%	0.402
frame	MobileCLIP	Dev 5% FPR	0.634	16.12%	20.28%	0.183
frame	SigLIP	Max dev F1	-2.611	7.02%	22.92%	0.080
frame	SigLIP	Dev 5% FPR	0.491	0.00%	1.67%	0.000
window	MobileCLIP	Max dev F1	0.589	76.03%	65.83%	0.409
window	MobileCLIP	Dev 5% FPR	0.650	57.85%	35.14%	0.441
window	SigLIP	Max dev F1	-2.686	55.37%	30.00%	0.453
window	SigLIP	Dev 5% FPR	-0.377	7.02%	5.14%	0.115

An empirical 5% development constraint does not ensure 5% false positives on another camera.

Detecting an interval can still mean many alarms

Screen: at least one raw categorical alarm inside each reviewed source/label interval

Input	Model	Positive intervals hit	Normal intervals with alarm
frame	MobileCLIP2-S2	5/5	7/7
frame	SigLIP2-B/16-224	0/5	3/7
frame	SmolVLM-500M	5/5	7/7
frame	MiniCPM-V 4.0	3/5	3/7
window	MobileCLIP2-S2	4/5	7/7
window	SigLIP2-B/16-224	2/5	2/7
window	SmolVLM-500M	5/5	7/7
window	MiniCPM-V 4.0	4/5	1/7

Any-alarm counts depend on interval length and do not establish operational event detection.

The remaining datasets are usable as diagnostics

Reported values are target-prediction rates; verified event recall, precision and FPR remain unavailable

Candidate query	Inputs	MobileCLIP	SigLIP	SmolVLM	MiniCPM
Littering	6,667	0.76%	26.71%	74.86%	55.77%
Loitering	1,675	10.63%	84.54%	0.12%	4.60%
Rough sleeping	111	9.91%	95.50%	81.08%	66.67%

Littering: generic action boxes, no matched normals.
Loitering: person boxes do not establish duration.
Rough sleeping: lying/fall posture does not establish sleeping outdoors.
High prediction rates cannot establish event quality; low rates may reject unsupported claims.

Seeing a person does not establish loitering

Two definitions queried on the same 1,675 person-labelled development images

Model	Person-presence positive hit	Loitering target-prediction rate
MobileCLIP2-S2	28.54%	10.63%
SigLIP2-B/16-224	17.37%	84.54%
SmolVLM-500M	92.66%	0.12%
MiniCPM-V 4.0	96.12%	4.60%

SmolVLM and MiniCPM usually recognize people. Duration and event evidence still require temporal labels.

Oracle crops identify a promising component

Synthetic lying-versus-standing diagnostic; 44 lying + 4 standing test crops

Model	Precision	Recall	F1	FPR	Missed
MobileCLIP2-S2	63.64%	15.91%	0.255	100.00%	37
SigLIP2-B/16-224	100.00%	100.00%	1.000	0.00%	0
SmolVLM-500M	100.00%	59.09%	0.743	0.00%	18
MiniCPM-V 4.0	100.00%	90.91%	0.952	0.00%	4

SigLIP: 48/48 correct; MiniCPM: 40/44 lying + 4/4 standing. Only four controls; boxes are supplied.

Smoking remains a difficult task

Prior GPU evidence reused; image models and supervised offline VadCLIP have different protocols

Image model / dev	Recall	F1	FPR
MobileCLIP2-S2	0.56%	0.011	0.22%
SigLIP2-B/16-224	0.00%	0.000	0.00%
SmolVLM-500M	5.41%	0.095	6.14%
MiniCPM-V 4.0	0.00%	0.000	0.00%

VadCLIP: supervised offline

1,056 source-split test frames
AP 0.633; AUROC 0.587
Recall 51.8%; FPR 35.7%
F1 0.565; selected dual_mean

Different training and temporal context.

3,278 smoking development frames are already used. VadCLIP is a separate supervised smoking study.

Choose the next experiment by failure mode

No deployment winner across all five anomalies is established by these source datasets

Model	Useful observed evidence	Issue to resolve before deployment
MiniCPM	Best raw unattended frame F1	Misses 65% of positive endpoints
SigLIP	Strong oracle posture discrimination	Raw decisions / threshold transfer
MobileCLIP	High unattended endpoint recall	High false alarms; prompt/crop sensitivity
SmolVLM	High person-grounding hit rate	Near-universal alarms; 1,457 abstentions

Next: independently verify the event labels; add real littering, duration and sleeping events with matched controls; include fresh cameras. Evaluate detector crops and native temporal inputs as development experiments, then freeze a fresh test.

Updated results are traceable and reviewable

Raw inference artifacts are preserved; this deck uses the new expanded accuracy report

Results and source evidence

Expanded accuracy report + numeric CSV
40 model summaries + 55,156 predictions
Frozen CLIP thresholds
Reviewed source intervals + camera groups

Paths and source hashes are in the notes.

Validation and presentation

144 existing tests passed
40/40 metric recomputations matched
All runs: CUDA, unpaced, zero errors

Editable PowerPoint + vector PDF
No additional latency evaluation

Primary limitations: provisional event labels, two test camera families, weak event proxies and four pose controls.